

The Lever Generalizes — and It Brakes: A Late, Bidirectional Action-Commitment Lever Across Agent Decisions and Architectures

Toward mechanistic circuit-breakers for irreversible agent actions

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Abstract

A prior result (*The Lever Is Late*) showed that, for a long-horizon coding agent, control of the **finish** decision lives not at the mid-layer “task is done” verdict but in a late, task-matched action-commitment block ~ 30 layers downstream. That left two questions open: is the late lever *specific to termination*, and can it *brake* an action, not just elicit one? We answer both on Qwen3.6-27B over 99 SWE-bench Pro trajectories, using a second, distinct decision available in the same data — commit a file edit (**str_replace_editor**) versus continue reversible exploration (**bash**) — with $n=60$ deterministic decision points per condition, prefill-only patching, and generation-confirmed outcomes. **(1) Generalization (elicit).** At an explore decision, injecting a task-matched *edit*-donor into the late block makes the agent emit a real **edit** call: $0.23 \rightarrow 0.77$ at L59 (position control 0.08, cross-task control 0.48). **(2) The brake (suppress).** At a commit decision, injecting an *explore*-donor collapses the real edit rate $0.48 \rightarrow 0.02$ (96% suppression) at L55, with a same-class control intact (0.55) and the opposite-direction donor boosting to 0.92. **(3) The mechanism is monotonic and bidirectional.** Exact paired McNemar on all 14 per-point conditions yields seven contrasts that survive Holm-Bonferroni ($m=7$, worst $p=7.6e-5$); the discordance structure is the headline: elicit has $c=0$ (the edit-donor only *turns commits on*, never silences) and the brake has $b=0$ (the explore-donor only *turns them off*, never lights). The late lever moves exactly in the donor’s direction with $\sim$ zero off-direction noise. **(4) Cross-architecture.** The late-commitment *geometry* and a donor-specific late *writability* replicate in a different model family (Mistral-7B-Instruct-v0.3): the tool-commit decision is flat/negative through the first half and explodes in the late block, and a task-matched edit-donor raises $P(\text{edit})$ where a same-norm random vector destroys it. We frame the bidirectional late lever as the mechanism for a *mechanistic circuit-breaker*: a single late-layer intervention that blocks an action at its commit point. We demonstrate it on a state-mutating but *undoable* edit; intervening on a genuinely irreversible action (e.g. **send_transaction**) is the named next step. Tool, pre-registration, per-point data, and an adversarial pre-publication evaluation are released.

1 Background and contribution

The WANDERING arc established, across five studies on Qwen3.6-27B (reasoning-tuned, 27B parameters), a detection-control asymmetry for long-horizon agent failure: internal states that

predict a behavior routinely fail to *control* it. The arc’s first positive, *The Lever Is Late*, located control of the **finish** decision in a late action-commitment block (L51–L63), ~ 30 layers downstream of the mid-layer verdict, and only when steered with a *task-matched* donor. We released **decision-locator**, a model-agnostic tool implementing three primitives — LOCATE (logit-lens per layer), SWEEP_PATCH (donor injection per layer), and STEER_GENERATE (prefill-only patch of the decision position, then free decoding). This sits in the activation-steering / representation-engineering line [6, 7], applied to a long-horizon agent’s *action channel* across turns rather than a single-turn attribute.

This paper pre-registers and answers two questions that the single **finish** result could not (PREREG_commitment_lever.md, PREREG_cross_model.md). First, is the late lever a property of the *action-commitment channel* or an idiosyncrasy of termination? Second, can the lever *suppress* an action at its commit point — the mechanism a safety *circuit-breaker* would need? Our contributions:

1. The late lever **generalizes** from **finish** to a second committal action (commit an edit), by the same locate-patch-steer method, generation-confirmed (§3).
2. The same lever **brakes**: a late explore-donor patch suppresses a real commit by 96% (§4).
3. The effect is **monotonic and bidirectional** — elicit $c=0$, brake $b=0$ — with all seven pre-registered contrasts surviving Holm–Bonferroni on exact paired McNemar (§5).
4. The commitment **geometry and donor-specific writability replicate cross-family** on Mistral-7B (§6).
5. We frame this as a **mechanistic circuit-breaker** and state honestly what is shown (a semi-irreversible proxy) and what is not (a genuinely irreversible action) (§8).

2 Setup

Data and decision points. We reuse the public bundle of 99 SWE-bench Pro [8] trajectories from Qwen3.6-27B (success 40, locked 39, wandering 20). The agent’s action space is three tools: **bash** (3250 calls; reversible exploration), **str_replace_editor** (1101 calls; commits a file edit; semi-irreversible), and **finish** (40). A *commit decision point* is any turn poised between committing an edit and continuing exploration; we reconstruct each faithfully at the tool-choice position, deterministically (sorted traces, ≤ 3 /trajectory), giving $n=60$ edit-decision and $n=60$ bash-decision points. **Patching.** All interventions use the **decision-locator** prefill-only protocol: the late-position residual is replaced once at the decision token, then the model decodes greedily; patching every step degenerates into repetition and is not used. **Fidelity gate.** Before any causal number, $P(\text{edit})$ at reconstructed edit points (0.474) exceeds that at bash points (0.299), confirming the states reproduce the real commit propensity (decision turns 5.2 vs 4.5 — close, weakening a positional confound; the position-matched control below addresses it directly).

3 The late lever generalizes: eliciting a second action

At a **bash** (explore) decision point we inject a task-matched *edit-donor* late-block residual and ask whether the agent emits a real **str_replace_editor** call (Table 1). The baseline emission rate is 0.23; the task-matched edit-donor raises it to 0.77 at L59. The two controls isolate the effect: a *position* control (an explore-donor at the same points) stays at 0.08 — below baseline, ruling out “any late patch elicits an edit” — and a *cross-task* edit-donor reaches 0.48, well above baseline but below the task-matched donor. As in *The Lever Is Late*, task-matching adds a real increment; unlike

finish, the `edit` decision also carries a substantial *generic* component (cross-task $\gg$ baseline), so task-specificity is *graded per action*, not absolute.

condition (at bash decision points, $n=60$)	emission rate	exact McNemar vs. baseline
no patch (baseline)	0.23	—
task-matched edit-donor @L59	0.77	$4.7\text{e-}10$ ($b=32, c=0$)
task-matched edit-donor @L55	0.62	$2.4\text{e-}7$ ($b=23, c=0$)
position control (explore-donor) @L59	0.08	— (vs. donor: $9.1\text{e-}13$)
cross-task edit-donor @L59	0.48	— (vs. donor: $7.6\text{e-}5$)

Table 1: Elicit. A task-matched late edit-donor makes a stuck-in-exploration agent emit a real edit call.

4 The same lever brakes: suppressing a commit

At an `edit` (commit) decision point we inject an *explore*-donor and measure whether the real edit emission is suppressed (Table 2). The baseline edit rate is 0.48; the explore-donor at L55 collapses it to 0.02 — a 96% suppression. The controls are clean: a same-class (edit-donor) patch at L55 does *not* suppress (0.55), and the opposite intervention — a task-matched edit-donor at L59 — pushes the rate *up* to 0.92. Thus L55 is the brake point and L59 the power point ($0.03 \leftrightarrow 0.92$; Figure 1). A generation-confirmed, single late-layer patch blocks a real commit.

condition (at edit decision points, $n=60$; baseline edit rate 0.48)	edit rate	exact McNemar
explore-donor (brake) @L55	0.02	$7.5\text{e-}9$ ($b=0, c=28$)
explore-donor (brake) @L59	0.03	$1.5\text{e-}8$ ($b=0, c=27$)
same-class control (edit-donor) @L55	0.55	—
opposite (edit-donor) @L59	0.92	—

Table 2: Brake. A late explore-donor patch suppresses a real commit; same-class control intact.

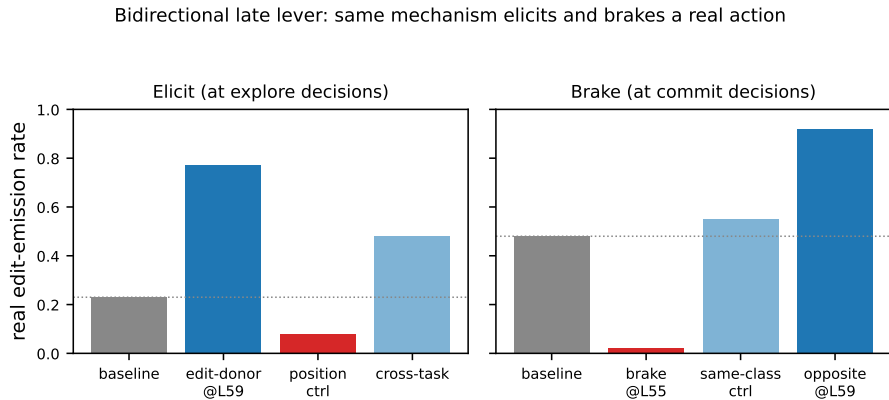


Figure 1: The same task-matched late lever elicits a real edit at an explore decision (left) and brakes a real edit at a commit decision (right), each against position / same-class and cross-task / opposite controls.

5 The mechanism is monotonic and bidirectional

We logged the per-point binary outcome for all 14 conditions and computed exact two-sided paired McNemar [10] tests. All seven pre-registered contrasts survive Holm–Bonferroni [11] ($m=7$; worst, task-specificity, $p=7.6e-5$). The discordance structure is the result: every elicit contrast has $\mathbf{c}=\mathbf{0}$ (the edit-donor turns commits *on*, $b=23\text{--}41$, and *never* silences one that already fired), and every brake contrast has $\mathbf{b}=\mathbf{0}$ (the explore-donor turns commits *off*, $c=27\text{--}32$, and *never* lights one). The late lever therefore moves the decision *exactly* in the donor’s direction with $\sim$ zero off-direction noise — a structural, not merely a rate-shift, claim. The one-token $\Delta P(\text{edit})$ profile corroborates: monotonic from L23 (-0.003) to L63 ($+0.685$), with mid layers (L23, L31) inert and the explore-donor control going *negative* late (-0.28) — the one-token signature of the brake (Figure 2).

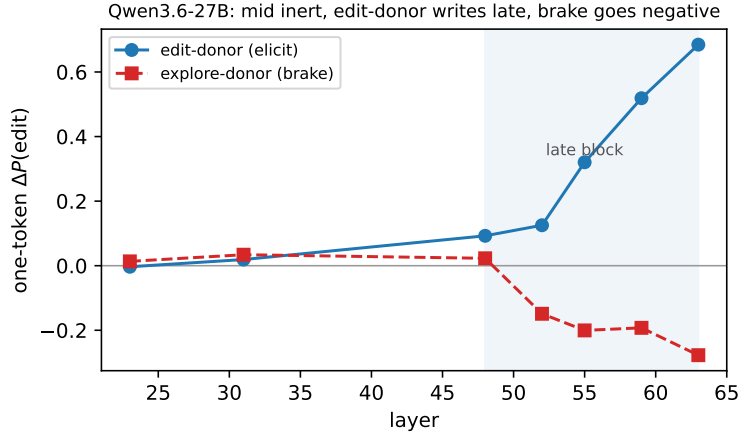


Figure 2: One-token $\Delta P(\text{edit})$ by layer (Qwen3.6-27B). The edit-donor (elicit) is inert mid-stream and rises monotonically to L63; the explore-donor (brake) goes *negative* in the same late block. The lever is late, and bidirectional.

6 Cross-architecture replication

To test whether the late-commitment geometry is architectural rather than Qwen-specific, we ran a lite locate-plus-one-token test on **Mistral-7B-Instruct-v0.3** (a distinct model family) over 30 SWE-bench Pro tasks, with context-defined edit- vs. explore-decision points. **Geometry (robust over three runs)**: the logit-lens edit gap is flat/negative through the first half (L1–L15, ~ -2) and explodes in the late block (L19 $+3.2$, L23 $+6.8$, L27 $+10.3$) — the same mid-flat, late-explosion shape as Qwen. **Writability**: injecting a task-matched edit-context donor into an explore decision raises $P(\text{edit})$ donor-specifically (e.g. $+0.15$ at L27), whereas a same-norm *shuffled* vector *lowers* it (-0.61 at L27); the donor-specific margin is $+0.6\text{--}0.76$ in the late block. The donor is inert at L9/L11 and writes from L13 on. The qualitative pattern — a late-emerging, direction-specific, written-not-destroyed commitment representation — replicates across families (Figure 3).

Scale-matched replication (Mistral-Small-24B). To retire the “7B-only” concern, we re-ran the same lite test on **Mistral-Small-24B-Instruct-2501**, scale-comparable to Qwen3.6-27B. The result is *cleaner* than the 7B and matches Qwen more closely. *Geometry*: flat/negative through L17, exploding L19→L37 ($+1.8 \rightarrow +6.4$). *Fidelity*: $P(\text{edit})$ is 0.955 at edit-context versus 0.007

at explore-context — a far sharper context separation than the 7B (0.84 vs 0.74) and comparable to Qwen. *Writability — clean mid-vs-late dissociation*: the edit-donor is *inert in the mid block* (L11/L13/L15: $-0.003 / -0.002 / -0.005$) and *writes almost fully late* (L29–L39: $+0.95$ to $+0.98$), donor-specific (edit-donor $+0.97$ vs. shuffled-neutral $\sim +0.5$). This is the *exact* mid-inert / late-write shape of the Qwen Tier-1 result — sharper than the 7B (which wrote from mid-depth). The late lever thus replicates across two families *and* two scales (Table 3), with the mid/late dissociation cleanest at the scale-matched 24B (Figure 4).

	Qwen3.6-27B	Mistral-7B	Mistral-24B
geometry: late-explosion	L51–63	L19–27	L19–37
fidelity $P(\text{edit})$ edit/explore ctx	0.47 / 0.30	0.84 / 0.74	0.955 / 0.007
writable window onset	late-only	mid ($\sim$ L13)	late-only (mid inert)
donor-specific late margin	gen-confirmed	$+0.6$ – 0.76	$+0.97$ vs neutral

Table 3: The late commitment lever across two architecture families and two scales. The scale-matched 24B reproduces the Qwen mid-inert / late-write dissociation most cleanly.

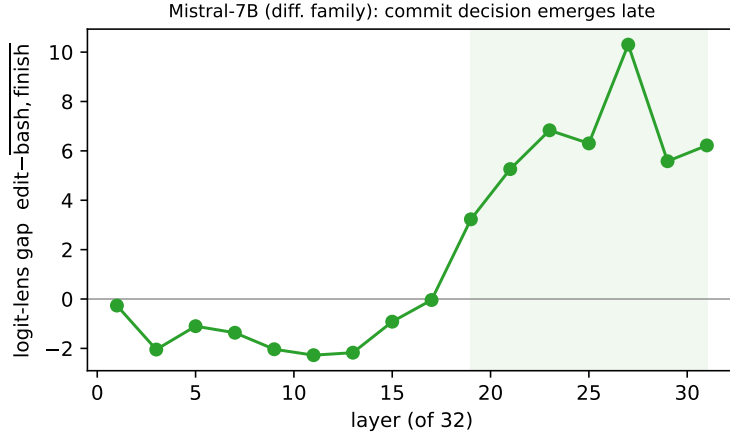


Figure 3: Cross-family LOCATE (Mistral-7B-Instruct-v0.3). The logit-lens commit-vs-explore gap is flat/negative through the first half and explodes in the late block (shaded) — the same late-emergence geometry as Qwen3.6-27B, in a different architecture.

7 Opening the lever: a sparse attention-head circuit

The lever is a single late-layer residual patch; we now open it. Because a pre-norm block writes its output as an *exact* sum, $y_L = x_L + a_L + m_L$ (input residual, attention contribution, MLP contribution), the published donor patch $\Delta_{\text{full}} = y_L^{\text{donor}} - y_L^{\text{point}} = \Delta x + \Delta a + \Delta m$ is exactly additive in the residual. We therefore re-inject each component alone with an additive hook at the layer- L output and read the one-token $\Delta P(\text{edit})$, the same metric as the dense profile of §5. Two gates hold throughout: the additive split reconstructs the captured residual to relative error 0.0025 (L55) / 0.0024 (L59), and the **full** re-injection reproduces the published lever exactly (elicit emission $0.23 \rightarrow 0.77$ @L59, brake $0.48 \rightarrow 0.03$ @L55), so the decomposition measures the same intervention. All numbers are $n=60$ /condition on Qwen3.6-27B.

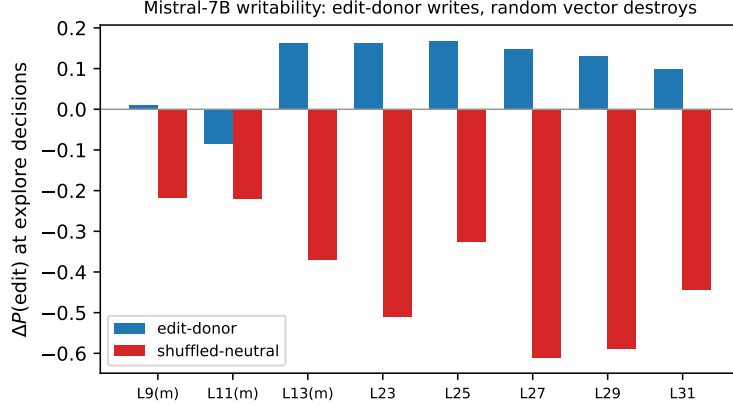


Figure 4: Cross-model writability. The task-matched edit-donor (solid) is inert mid-stream and writes the commit decision in the late block, while a same-norm shuffled-neutral vector (dashed) destroys it everywhere — the mid-inert / late-write dissociation, cleanest in the scale-matched Mistral-Small-24B.

Attention writes the elicit; the MLP is inert; the brake is distributed. Table 4 gives the recovery fraction $r = \Delta P_{\text{comp}} / \Delta P_{\text{full}}$ of each component. The two directions decompose *differently*. For the **elicit**, the L59 attention sublayer is the single largest writer ($r_{\text{attn}}=0.43$, 95% CI [0.33, 0.53]), the **MLP is null** ($r_{\text{mlp}}=-0.04$; paired Wilcoxon $\text{attn} \gg \text{mlp}$ $p=1.8\text{e}-8$), and the attention write is *direction-specific* (task-matched $\Delta P=+0.223$ vs. random-donor $+0.100$, $2.2\times$); under generation the attention component alone roughly doubles emission ($0.23 \rightarrow 0.45$) while the MLP component is inert (0.25). At L55 neither sublayer writes the signal — it is still *arriving* via the residual from below ($r_{\text{inp}}=0.52$) — so the attention write fires specifically at L59. The **brake** localizes to *neither* sublayer ($r_{\text{attn}}, r_{\text{mlp}} \approx 0$ at both layers); $\sim 35\text{--}40\%$ is carried by the upstream residual and the remainder by a large negative *nonlinear interaction* (the gap $\Delta P_{\text{full}} - \sum \Delta P_{\text{comp}}$ is $-0.23 / -0.26$). The MLP is null in all four cells. This rules out a Geva-style feed-forward “emit-action” key-value write [14]: the late commitment is an attention/residual phenomenon, and the elicit/brake asymmetry — attention-written push vs. distributed, irreducible suppression — holds at sublayer resolution (Figure 5). The strict pre-registered threshold ($r_{\text{attn}} \geq 0.5$) is not met, so we state it honestly: the elicit is attention-*dominated* but not attention-*sufficient*.

direction	layer	ΔP_{full}	r_{inp}	r_{attn}	r_{mlp}
elicit	L55	+0.319	0.52	0.05	-0.09
elicit	L59	+0.518	0.22	0.43	-0.04
brake	L55	-0.404	0.40	0.02	0.03
brake	L59	-0.382	0.35	0.00	-0.02

Table 4: Attn-vs-MLP decomposition (recovery fraction $r = \Delta P_{\text{comp}} / \Delta P_{\text{full}}$). The elicit is attention-written at L59 (MLP null, direction-specific); the brake localizes to no sublayer.

The elicit is a sparse three-head push circuit. L59 is a full-attention layer in Qwen3.6’s hybrid linear/full stack (every fourth layer is softmax attention), with 24 query heads. The attention output is again an exact sum over heads, $a_L = \sum_h c_h$ with $c_h = z_h W_O^{(h)}$ [15], so we re-inject each

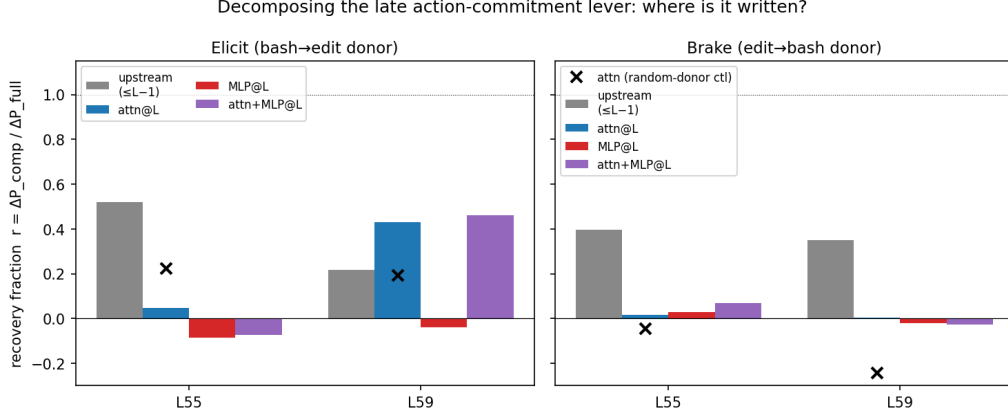


Figure 5: Component recovery of the late lever. Elicit (left) is carried at L59 by the attention sublayer (blue), MLP (red) null; brake (right) by neither sublayer. The MLP is inert in all four cells.

head’s donor→point difference Δc_h alone (head-split reconstruction relative error 0.0015). The result is *sparse*: heads 8, 6, 3 carry +0.122, +0.119, +0.083 while the rest are ≈ 0 or negative, and the top three together ($\Delta P = +0.262$) *exceed* all 24 heads jointly (+0.224) because an opposing set (16, 19, 21, ..., each -0.01 to -0.03) partially cancels them — a small push–pull circuit, not a diffuse sum (Figure 6). Behaviorally the three heads alone lift emission $0.23 \rightarrow 0.42$ ($\sim 89\%$ of the full attention channel’s 0.45), and they are direction-specific (head 8 task-matched +0.121 vs. cross-repo +0.077). Geometric write-magnitude *misleads*: the largest writer along the attention direction (head 21, projection 14.0) is causally an *opponent* (-0.014) — only the patch tells the truth. The brake, re-scanned per head, has *no* head locus and is not a cancellation either (all 24 heads in $[-0.005, +0.012]$), so the elicit/brake asymmetry holds all the way down to head resolution.

head set (L59, elicit)	#heads	$\Delta P(\text{edit})$	edit-emit
top-1 {8}	1	+0.121	—
top-3 {8, 6, 3}	3	+0.262	0.42
all heads	24	+0.224	0.47
top head, cross-repo donor (ctl)	1	+0.077	—

Table 5: Cumulative head contribution. Three heads reproduce (and overshoot) the full attention effect; emission baseline is 0.23. Top-3 > all-24 because an opposing head set partially cancels the push.

What the heads read, and a causal source test. At the decision token the commit heads attend *globally* (low locality, ~ 0.10 – 0.16 of mass in the last 128 keys) to the trajectory’s *tool-call history* — the tool names `bash` and `str_replace_editor` and the call’s JSON scaffold — not to any semantic “task done” verdict. This is an induction/copy signature [16]: the late decision promotes a tool name from the pattern of prior calls. We tested necessity by ablation [17]. The ideal head-edge knockout (recomputing $o_h = a_h V_h$) is blocked here: Qwen3.6’s full attention is *output-gated* ($z_h = g_h \odot (a_h V_h)$); the ungated reconstruction fails, relerr 1.7–2.4), so we instead ablate the *source content*, mean-patching the L59 key/value at the tool-name positions. The agent’s tool choice is then causally specific to each tool’s name tokens: ablating `bash` tokens drops $P(\text{bash})$ by -0.071

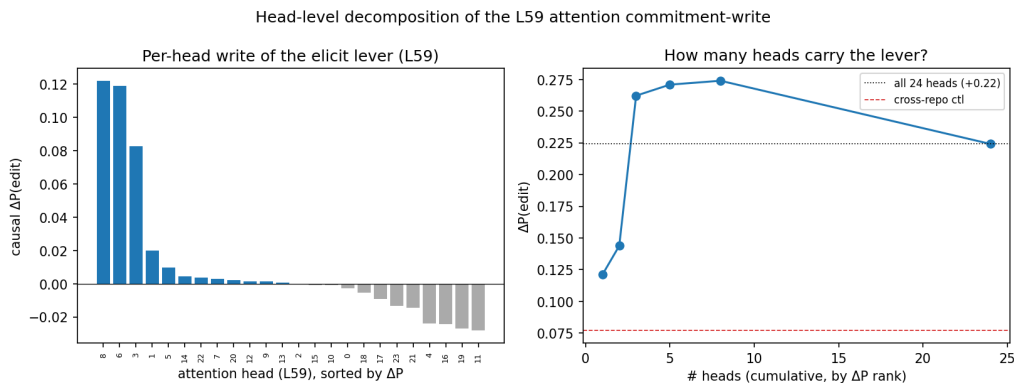


Figure 6: Head-level decomposition at L59. Left: per-head causal $\Delta P(\text{edit})$, sorted — a few heads push, a few oppose. Right: cumulative recovery; three heads reach (and exceed) the all-heads net.

(−18% relative) versus −0.000 for random, count-matched positions; ablating `str_replace_editor` tokens lowers $P(\text{edit})$ and raises $P(\text{bash})$. Honestly, the edit side is *ceiling-confounded* — the rows that contain `str_replace_editor` tokens are already edit-saturated ($P \approx 0.9$), so its drop is small (−0.031) — and this ablation is over all heads, not head-specific. The copy reading is therefore *supported, not nailed*; a gate-aware head-edge knockout is the named next step.

Synthesis. Four levels down, the late action-commitment lever resolves to a concrete, sparse circuit: a handful of induction-style attention heads in a single late full-attention layer (L59) that read the trajectory’s tool-call history and promote a tool name, partially opposed by a counter-set — not a feed-forward feature and not an attention to a semantic verdict. The brake, by contrast, has no sublayer and no head locus: it is a distributed, super-additive residual effect. The agent decides whether to commit an action largely by *copying the pattern of tools it has already used*.

8 Discussion: a mechanistic circuit-breaker, and its honest scope

The bidirectional late lever is the mechanism a safety *circuit-breaker* requires: a single intervention at an action’s commit point that can block it. We emphasize what is and is not shown.

- **Generalization is an existence proof, not a universal law.** It holds for `finish` and one second action, on one model and one task family (plus a cross-family geometry/writability replication). More actions, models, and task families remain.
- **The proxy is semi-irreversible.** `str_replace_editor` mutates state but is undoable. The genuinely irreversible case (`send_transaction`, `delete`) is untested and is the named next step; the circuit-breaker is demonstrated *on a proxy*.
- **The brake redirects to exploration.** Under the brake, `bash` is the dominant action (0.60). We measured the no-brake bash baseline at edit points (0.43): the brake therefore *lifts* exploration by +0.17 above its already-present floor while collapsing the commit, so the suppressed decision is genuinely re-routed to reversible exploration rather than merely stalled.
- **Emission is validated by a full-call re-parse.** The elicitor rate counts the tool-name prefix in a greedy continuation; to retire the onset-only concern we re-parsed the four headline conditions for the *full* `str_replace_editor` call. The lift survives in valid calls: elicitor 0.23 →

0.37 (valid-call), brake 0.40 $\rightarrow$ 0.07. Onset is thus an upper bound but the direction holds on valid calls in both directions; the brake (suppression) is the more robust to this.

These caveats were produced by an adversarial pre-publication evaluation (EVAL_commitment_lever.md) that recomputed every number from the released per-point data (determinism confirmed) and corrected two earlier overclaims; we release it alongside the paper.

Relation to prior work. This extends activation steering / patching [6, 7] from single-turn representations to a long-horizon agent’s action channel, and operationalizes the knowledge–action gap as a *layer* gap with a *bidirectional* late lever. Our use of *circuit-breaker* is a safety-interlock metaphor and is methodologically distinct from the training-time *Representation Rerouting* of Zou et al. [12], which fine-tunes a model so harmful *content* representations become incoherent; we instead apply a single *inference-time*, late-layer patch at an agent’s *action*-commitment point, with no fine-tuning. The brake direction is related to the refusal direction [13] but the steered quantity is the agent’s *action commitment* (redirecting a committal action to a reversible one) rather than a generic refusal. The result is consistent with the arc’s detection $\neq$ control thesis and with mechanistic-corrigibility framings of pausing before irreversible actions.

9 Conclusion

The late action-commitment lever is not specific to termination and is not one-directional: a single task-matched late patch both elicits and brakes a real agent action, monotonically and donor-specifically, with a geometry that replicates across model families. This is the mechanism for a mechanistic circuit-breaker. The next step is to demonstrate it on a genuinely irreversible action and at scale-matched second models.

Artifacts. `decision-locator` (model-agnostic), the pre-registrations, the per-point data and deterministic decision-point state, the exact-statistics script, and the adversarial evaluation are released at openinterp.org/research and on the public data bundle.

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